

Springfield High School

Physics (Grade 10) — Homework Assignment

Name: _____

Roll No: _____

Class: _____ Section: _____

Date: _____

Time: 40 min

Max Marks: 40

Instructions:

- All questions are compulsory.
- Draw diagrams neatly with a pencil.
- Show all steps in numerical problems.

Q1. State Newton's second law of motion and write its formula.

[3]

Ans. It states that force equals rate of change of momentum.

Formula: $F = m \times a$

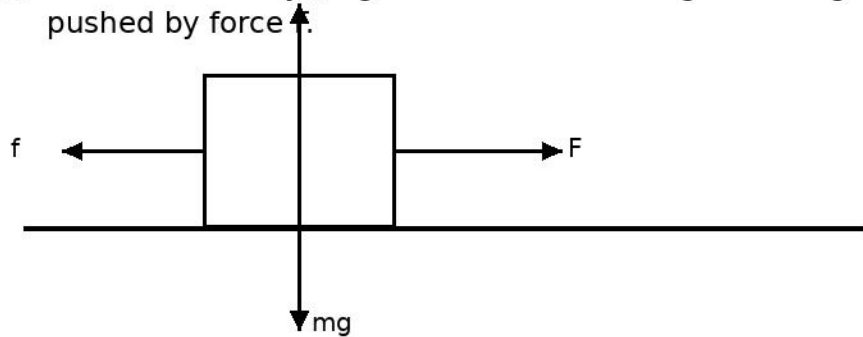
Q2. A body of mass 5 kg accelerates at 3 m/s^2 . Find the force.

[3]

Ans. $F = m \times a = 5 \times 3 = 15 \text{ N}$

Q3. Draw a free-body diagram of a block resting on a rough horizontal surface being pushed by force.

[5]



Q4. Define kinetic energy and give its SI unit.

[2]

Ans. Energy possessed by a body due to its motion. SI unit: joule (J).

Q5. Complete the motion table for uniform acceleration $a = 2 \text{ m/s}^2$, starting from rest:

[5]

t (s)	v (m/s)	x (m)
0	0	0
1	2	1
2	4	4
3	6	9

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- Q6.** Calculate the pressure exerted by a force of 100 N on an area of 5 m². **[3]**
Ans. $P = F / A = 100 / 5 = 20 \text{ Pa}$
- Q7.** Using Ohm's law, find the current when $V = 12 \text{ V}$ and $R = 3 \text{ ohm}$. **[3]**
Ans. $I = V / R = 12 / 3 = 4 \text{ A}$
- Q8.** Explain the term 'inertia' with one everyday example. **[4]**
Ans. Inertia is the tendency of a body to resist a change in its state of motion.
Example: passengers lurch forward when a bus stops suddenly.
- Q9.** Write the formula for momentum and calculate it for $m = 3 \text{ kg}$, $v = 4 \text{ m/s}$. **[3]**
Ans. $p = m \times v = 3 \times 4 = 12 \text{ kg.m/s}$
- Q10.** State the law of conservation of energy. **[4]**
Ans. Energy can neither be created nor destroyed, only transformed from one form to another.
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— End of Physics Paper —